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Learning History Through Computer Game Authoring

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Abstract— Recent researches showed that computer game authoring is able to help students to acquire mathematics and science knowledge in the classroom. The researchers believe that this method is able to apply for the other subject matters. The subject matter that the researchers focused in this paper is history learning. Firstly, this paper describes the process of game authoring workshop for history learning environment. Next, we analyzed participants' perspectives through interviews to seek for the learning experiences of using this method in history learning context. Lastly, we look into the feasibility of creating 2D computer game environment to support history learning and the participant's experience towards the authoring tool being used in this study. There are five participants in this workshop; all the participants have no prior experience in computer game authoring. In most of the approaches, students use computer game authoring tool to create 3D computer game; but in this workshop, the authoring tool being used is to create 2D Role-Playing Game (RPG). This is to reduce the rendering process during the workshop. In this case, RPGMaker is the authoring tool that being used by participants to create RPG.

Index Terms— computer games authoring, history learning, constructionism, computer games learning

I. INTRODUCTION

The rapid growth of computer games industry thus stimulates the use of computer games in teaching and learning. Youth nowadays expose to computer games almost every day through the game consoles or the computers. For example, a study that carried out in 16 secondary schools in Malaysia, 92.1% of them have experienced in playing video games [1]. Another survey in UK reported that 53% of 11 to 14 year olds play games four times a week or more, and that 44% play for more than one hour at time [5]. They are familiar with games, and hence educators nowadays are trying to immerse students into the subject matter through computer games.

Teaching history subject has been a challenging task for school's teachers. In students' perceptions, history is difficult and boring because there are too much of facts, time, and

historical events to be memorized [7]. Learning history is important for the young generation because it has high cultural values. Thus, it is important to find alternative yet interesting ways for young generation to study history.

This paper goes beyond the state of students learn the subject matter by playing computer games, researchers were considering the benefits of computer game authoring as the medium to enable history learning process with the references to previous research. According to Lim (2011), computer game authoring is able to instill history thoughts to the learners. Hence, this study is focused in designing a game-authoring workshop of history learning based on the proposed framework [4]. The game authoring tool that would be using in the workshop is *RPG Maker 2003* by Enterbrain, Inc (available at: <http://rpgmaker.net/engines/rm2k3/>). The computer game authoring process, including the research findings from participants, are explained in this paper. Then, this research will further to explore the experiences of young generation towards learning history through computer game authoring. At the end of the game authoring session, participants were given their experiences in using *RPGMaker 2003* to author a historical RPG. There are positive feedbacks and negative feedbacks collected through the interview. The paper concludes with suggestions for future work.

II. RELATED WORKS

In the effort to bring the computer games authoring into educational environment, a group of researchers had conducted a Game Maker Workshop attended by ten secondary pupils (aged 12-15 year's old) [6]. The aims were to give opportunity to young people to tell their own stories through computer games and to explore the use computer games authoring as a method for developing literacy skills. The toolset being used in this workshop was *NeverWinter Nights*, which enables non-expect to create 3D virtual environment role-playing games.

The intention of this workshop was to improve the writing skill of the learners by using computer game authoring. It was based on the constructionism approach which learners were engaged with the context of story while creating the computer game for learning. The workshop carried out for a week and with a series of activities. The result showed that most of the

participants enjoyed the learning method, and the participants were more engaged with the creating the characters for storytelling. Besides, this workshop also found that the self-esteem of the learner could be improved during game making process.

Another project conducted by Kafai (1995) and his fellow researchers was focused on learning design by making games [2]. The particular focused in this project was on how students as designers integrate planning and problem solving while building an artifact. A class of 16 fourth-grade students was engaged during an extended period of time (6 months) in the design and production of educational games that taught fractions. The students who participated in this project were rarely using educational software; instead, they are working with the *LogoWriter* to create their software.

In this Project Headlight, the process of game development for 16 student designers was observed every day over the course of six months. A combination of qualitative methods was used to document the student's ideas, thoughts, and progress in the game development. Pre- and post-interviews were conducted to gather information on students' knowledge of programming. The researchers found that there was no uniform way in which students handled the early days of creating their games, or furthermore, reinforce the impression that there is no one "right way" to start a design task, and that many of students' choices in approach and content were related to their personal preferences. Students used approaches of both planning and bricolage in order to solve the task in a successful manner. From the project, students learned not only through design, but also about design, and reached a level of reflection that went beyond traditional school thinking and learning.

Learning by computer game authoring in school-aged children has covered a wide range of study. The topics covered include general science, social studies, mathematics, and environmental studies. A study had been carried out to examine child-invented games as possible educational materials for children and as a potential diagnostic tool for content, understanding, and also cognitive/learning gaps related to the dengue problem [3]. In this study, an eight-years-old Filipino American child was asked to draw and describe two games related to dengue fever. The child had prior knowledge about dengue. The case study took place in the child's home.

In the first game, the child was asked to create a game about dengue fever by papers and pencils. The child completed the game design around 30 minutes. After the completion of the game, the child was asked to describe the game, its play, and its rules. One day later, the child was asked to create the second game. The study found that the child was further instructed and asked more about dengue's infection and cures during the process of creating the second game.

Result showed that most of the game instructions created by the child were orally given. Thus, the game board design may have been more important to the child. Both games revealed the child's content understanding about dengue. These content topics reflected on the child's understanding about dengue

transmission and dengue-carrying mosquito control measures. Most of the child's information was correct. However, some conceptual issues needed correction.

There is limited number of research about the feasibility of game authoring tool to be used in history learning. However, the previous researches have given strong guidelines for the researchers to extend the Role Playing Game (RPG) authoring method in learning history. From the researches mentioned above, the toolset being used plays an important role in this learning environment in order to create enjoyable learning experience for the learners. Hence, the toolset should be non-intrusive, easy to navigate, and high in efficiency.

III. RPGMAKER TOOLSET

RPGMaker is a game-authoring tool that allows users to create turn-based RPGs. The first Windows-based RPGMaker was released in 1995. In this game authoring workshop, RPGMaker 2003 has been chosen as the authoring tool because it enhanced the functionalities and the user interface. The drag-and drop interaction has made the game authoring session become easier.

RPGMaker 2003 prepares a list of tile set (also known as Chipset for version before RPGMaker 2003) for users to create maps in the RPG. User chooses suitable tile set from the list, and then design the map layout freely. Apart of default tile set, users are able to download free tile sets from the Internet.

RPGMaker uses simple scripting for event handling. There is a proper user interface for users to select specific action from the Event Command List. Users are allowed to change the parameters of the command that they have selected to fit with the game environment. Figure 1 illustrates the Event Command List in RPG Maker.

As a conclusion, RPGMaker is a simple tool that can be used to create RPG without much knowledge in programming and mathematics calculation. In this study, participants would provide feedbacks on the capabilities and usability issue of the RPGMaker to support history learning context. The feedbacks are hoped to shape a new system's design and user interface for computer games authoring tool.

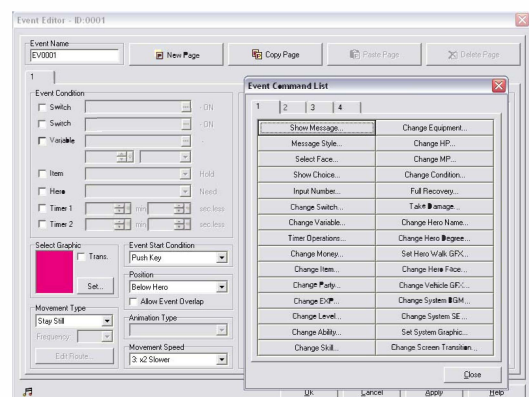


Fig.1. Event Command List in RPGMaker

IV. THE RPG AUTHORING WORKSHOP

The workshop was planned according to the conceptual framework for learning history using constructionism approach [4]. A group of five college new faces were invited to participate the RPG authoring workshop. They voluntarily took part in this workshop because they are enthusiastic in game authoring. However, they did not have any experience in computer game authoring. The group consists of four males and one female.

The workshop was carried out one session per week. There were 8 sessions in total with 60-75 minutes for each session. At the beginning of each session, participants were given time to discuss about their mission. This is to give directions for the participants on what they wanted to achieve. Then followed by RPG authoring according to their mission of the session. At the end of the workshop, they presented their RPG to other participants and discuss about their findings toward the historical events during the workshop.

In this study, the researchers would like to address:

- the viability of using RPGMaker as the authoring tool to create historical RPG in fostering history learning.
- the user experience of using RPGMaker to create historical RPG.

A. Workshop Setting

The workshop involved a series of activities as showed in Fig. 2. The participants were informed about the aims of this workshop. At the initial stage, the participants played 2D RPGs that were developed by RPGMaker. This is to build up the fundamental concepts and the functionalities of the RPGMaker.

1) Group discussion

The researchers shaped a focus group to discuss about type of games that they enjoy playing. The initial discussion was to bring all the participants together and create a comfortable environment for them to communicate. The outcome of this session was to identify the common historical topic that they wanted to explore.

2) Introductory to RPGMaker

The researcher prepared a series of tutorial to guide the participants in using RPGMaker. The purpose was to give the participants an overall idea about the technology that they would use to create the historical event that they planned in the previous stage. Then, the participants tried to discover the other functionalities of RPGMaker at their own pace. The researcher was available throughout the workshop to help the participants whenever they encountered problem in using the RPGMaker.

3) Subject matter study

In this phase, the participants were looking into the content for their RPG. They used various sources in finding the information. These include the Internet, textbooks, and videos. Then, they were requested to establish a storyline for their RPG based on the findings. At this point, the participants possessed with the basic idea about RPGMaker and were able to justify how to develop their storyline in the

RPG environment. They sketched the storyboard on the paper based on their findings and the requirements of each scene.

They shared and discussed about their findings at the end of the workshop. The interaction between participants led to the different perspectives towards the same historical event among the group. Hence, it created an arena to instil the historical thought to the participants.

4) Characters and game design

From the study that has been conducted in the previous phase, participants then require to design the game characters involved in their RPG. A short description would be provided by the participants which to describe the attributes of each character. The descriptions could be their motives, background information, characteristic, and mission in the RPG. The participants would also considered the cultural tools and norms in the past while designing their characters.

5) Plot planning

The participants continued to work on their story planning. Each participant started drawing the game flow in a blank sheet of paper. Then, they determined the protagonists and the antagonists of their RPG. Then, they exchanged thoughts and information within the group. They planned when and how the antagonist appears in the game and the time flow of the game. While creating the game flow, the participants found that there are “missing parts” in between the events. Hence, they have added own perspective to connect the missing parts and make the gameplay more fluently.

6) RPG authoring

Participants explored the functionalities of the RPGMaker at the beginning stage of the workshop. They tried to create simple game using RPGMaker before working on their historical RPG. They were able to master the basic control within a short period. Then, they proceed to learn the advanced functions in RPGMaker. These include auto-trigger for introductory of the RPG, interactive dialogues, changing the splash screen, manipulate the treasure chest, and the event-handling in the RPGMaker.

When the participants mastered the tool, they started to create their RPG. In spite of the obstacles they were facing during game authoring, they resolved the obstacles by helping each other during the workshop. At the end of the session, they played each other's RPG when it was ready. The game authoring phase spent the most time in this workshop.

7) Presentation

At the final stage, the participants would present their RPG in a presentation session. The presentation was divided into two parts. The first part was focused in presenting the RPG and the outcomes of the workshop. The second part of the session was an interview session where the participants share their experience and reflection in using RPGMaker as the authoring tool creating historical RPG.

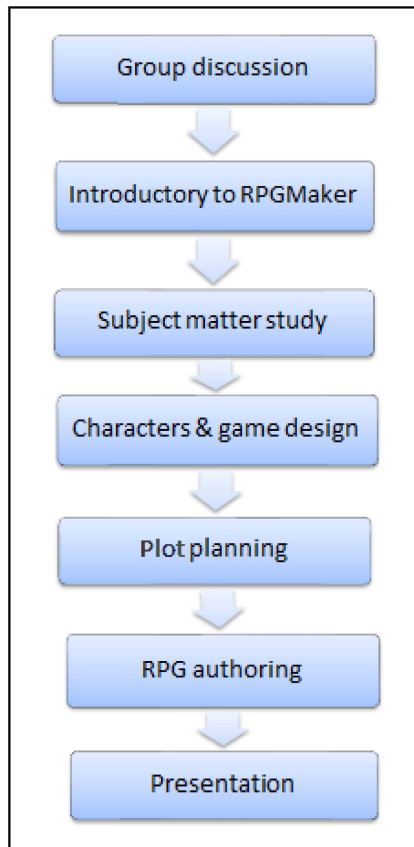


Fig.2. List of activities in RPGMaker workshop

V. RESULT

There were four out of five participants completed the RPG at the end of the workshop. Despite one participant did not complete the RPG but she attended the presentation and discussion sessions at the end of the workshop.

There is an important issue was reported by the participants regarding the workshop activities. Participants mentioned that they were asked to plan the game's plot before authoring the RPG using traditional approach, which was drawing the storyboard for their game play. However, it was not able to reflect interactive dialogues and the battle mode that would appear in the RPG. Besides, most of the participants changed the plot that was planned in advanced to accommodate the RPG scenario. This was because they had more ideas to contribute into the RPG during the authoring. However, the planning stage had given them a clear direction to start.

A. RPG Authoring in Learning History

Computer game authoring is a complex task. It involves mathematics calculations and programming skills in the past. Recently, releasing of computer authoring tools that allowed non-expert to create their own computer games without professional programming skills has changed the public perceptions over computer games authoring. This is because with these authoring tools, we are able to create computer games a lot easier. By looking at the popularity of computer games among the young generation, the researchers hope to engage students in history learning by applying RPG authoring

method. Hence, finding a suitable toolset is an important task to align with this purpose.

Based on the open-ended discussion being held at the end of the study, there were numbers of positive feedbacks from the participants. Table I showed the sample questions being asked during the discussion. The advantages of learning history using RPG authoring are to promote the group work within their peers, more engaging to the subject matter, and it is an innovative way that they have never experienced in the past. However, the participants suggested more time are needed for the sessions in order to complete the RPG. Table II showed the sample comments by the participants.

The end product of this workshop was a 2D historical based RPG created by the participants. The initiative of this project is to study the leverage of game authoring technology presents as new approach for constructionist learning environment for history subject. However, there are few issues arise by the participants during the workshop. Firstly, two participants mentioned that they preferred to create a 3D RPG than a 2D RPG. This is because 3D RPG is more appealing and more realistic. Secondly, the low resolution game display produced by RPG Maker did not appeal to the participants and somehow lowered their motivation in creating the RPG. This has showed that visual appealing of RPG plays an important role in fascinating the learners in the learning environment.

TABLE I. SAMPLE QUESTIONS

- 1) Do you see any difference between the traditional method and using RPG authoring in learning history?
- 2) Do you like this learning style to learn history?
 - a. If yes: What are the elements that you like?
 - b. If No: What are the elements that you dislike?
- 3) Is RPG authoring method able to promote your interest in learning history?
- 4) Based on your experience in this workshop, what are the advantages of applying this method in learning history subject?

TABLE II. SAMPLE COMMENTS

Topic	Sample comments
Positive feedbacks	I like the latter method; it makes learning history become more <u>enjoyable</u> .
Engaging the learners	I am more <u>engaged</u> to the subject being learned. I see it is an <u>innovative</u> way in learning history. It could practically engage me to the topic that I have to learn.
Improvements needed	I am fine. But it would be better if I have <u>more time</u> .

B. User Experience in RPGMaker

All the participants agreed that RPGMaker has the capabilities and useful for creating historical RPG. They highlighted the convenience of creating maps in RPGMaker. The simple click-and-draw method that allowed them to create a map in short time. The division between lower-tile layer and upper-tile layer in tile set gave a better insight in designing the game map. RPGMaker prepares the function that allowing users to import tileset which enable users to setup the environment of the story based on the specific historical events. The participants were tried to design the RPG environment according to their finding.

Most of the contents in history subject are delivered in text. RPGMaker prepares an easy way to create interactive dialogues for the game characters. Thus, participants were able to transform the historical facts into game play with simple steps. To achieve this, participants were required to gather data from various sources and then translate it into game dialogues. The convenience of RPGMaker acted as catalyst to make the process more interesting and more efficient.

Participants highlighted a useful feature in RPGMaker which is “*Change Weather*” function. This function allowed the participants to create different type of weather according to their needs in the game. The participants intended to change the weather based on the progression of the story. One participant claimed that weather changed during the game play able to make the story more realistic and can match with the climate of the historical events. In this case, RPGMaker supports user’s climate opinion in creating a historical game.

Combat mode in RPGMaker is easy to use, but participants claimed that the villains in the default setting were irrelevant to the real world. The “monster” or “vampire” as the villains in a historical RPG is not appropriate. Hence, the participants have imported relevant character sets to RPGMaker in order to fit to the real world circumstances.

As conclusion, the feedbacks from the participants about RPGMaker to be used as a authoring tool to create historical RPG are positive. They were able to complete most of the plots that they planned at the end of the workshop. Table III depicts the features of RPGMaker that voted by the participants based on the usefulness in creating RPG.

TABLE III. THE USEFULNESS OF RPGMAKER FEATURES

Task	Control method	Usefulness
Creating Map	Click-and-draw	
Changing Tileset	Import to library	
Assigning Dialogue	Type message directly to Event Command dialogue	
Making selection	Choose from Event Command dialogue	
Triggering combat mode	Choose from Event Command dialogue	

VI. CONCLUSION

This study is focused on the processes and the authoring tool for game authoring workshop in learning history. Through the study, the researchers gained deeper insight on the potential benefits of using computer games authoring in learning history. The findings of the RPG Authoring Workshop showed that young people find more enjoyable, engaging, and create new experiences in learning history. It has been demonstrated that there is a demand for learning by game authoring as the preferred method compared to traditional method of learning history. However, the result also showed that there is a need to amend the conceptual framework to accommodate changes during the RPG authoring phase. Last but not least, based on the findings, it is well worth to explore how to incorporate this new method in classroom curriculum to motivate students to study history.

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